
Seminar 13 - Solution

Review questions:

- Assume three firms: $MC_1 = 200, MC_2 = 200, MC_3 = 205$. The market demand function: $D(p) = 300 - p$.
 - How much will be the market price?
A: $p = 200$. Firms 1 and 2 push the market price towards their marginal costs.
 - How much will be output of Firm 1?
A: $y_1 = 50$. Total output ($y = 100$) is divided equally between Firms 1 and 2. Firm 3 is not able to produce anything for $p = 200$ (to compete with Firms 1 and 2) so its output is zero.
- Assume three firms: $MC_1 = 200, MC_2 = 203, MC_3 = 205$. The market demand function: $D(p) = 300 - p$.
 - How much will be the market price?
A: $p = 202$ (or just below 203 if you assume non-integer prices). Firm 1 has no incentive to push price down towards its marginal costs. It will undercut marginal costs of Firm 2 by setting price below 203. Thus, Firm 2 will leave the market. So Firm 1 remains the only producer and will be charging $p = 202$ (or just below 203).
 - How much will be output of Firm 1?
A: $y_1 = 98$. Firm 1 takes the whole market.
This is the profit-maximizing optimum. If you consider $p = 202, y_1 = 98$, profit is $\pi_1 = 202 * 98 - 200 * 98 > 0$. Do not forget about the total costs (!) when computing profit ($MC_1 * y_1$), we cannot just consider total revenue. On the other hand, profit is zero for $p = 200, y_1 = 100$ as price equals marginal costs.
So, the Bertrand's optimum for identical products when $p = MC$ assumes there are at least two firms with the lowest marginal costs.
- Assume two firms that compete on prices. The market demand function: $D(p) = 92 - p$; marginal cost functions: $MC_1(y_1) = 3y_1, MC_2(y_2) = 5y_2$. How much are the optimal outputs? How much is the market price?
A: $p = 60, y_1 = 20, y_2 = 12$. By setting the inverse demand function $p(y_1, y_2) = 92 - p_1 - p_2$ equal to MC_1 and then MC_2 and solving the equations for y_1, y_2 . Again, price equals marginal costs of both firms.

Exercises:

1. Consider two gas stations in the duopoly market where gas stations compete with each other with price of gas. Gas stations have following cost functions $TC_1(y_1) = cy_1 + 4,000$ and $TC_2(y_2) = cy_2 + 4,000$. Monthly demand functions are $D_1(p_1, p_2) = 1,000 - 30p_1 + 10p_2$ and $D_2(p_1, p_2) = 1,000 - 30p_2 + 10p_1$.
 - Due to high price of oil on financial markets were costs unit costs of gasoline in September $c^0 = 20$. Calculate Bertrand equilibrium (p_1^{b0}, p_2^{b0}) and corresponding outputs and profits.
A: $p_1^{b0} = p_2^{b0} = 32; y_1^{b0} = y_2^{b0} = 360; \Pi_1^{b0} = \Pi_2^{b0} = 320$
 - Decrease of oil price in October led to drop in unit costs of gasoline $c^1 = 10$. Because of low profits, firms decided to collude, therefore maximize joint profit $\Pi_1 + \Pi_2$ (no wealth redistribution) and choose corresponding prices p_1^k, p_2^k . Calculate these prices, outputs and profits of both firms.
A: $p_1^k = p_2^k = 30; y_1^k = y_2^k = 400; \Pi_1^k = \Pi_2^k = 4,000$
 - In November, both firms were punished for collusion by fine of 100 and restrictions for further collusion. Therefore, firms came back to Bertrand equilibrium, however with low prices $c^1 = 10$. Calculate equilibrium price p_1^{b1}, p_2^{b1} , outputs and profits.
A: $p_1^{b1} = p_2^{b1} = 26; y_1^{b1} = y_2^{b1} = 480; \Pi_1^{b1} = \Pi_2^{b1} = 3,680$.
If we want to compute profit in November, we should subtract the fine of 100 from this profit. However, if the collusion had not happened, 3,680 would have been the profit.

- Compare Π^k and Π^{b1} and decide if the collusion was profitable in spite of the fine 100.
A: Even with the fine is collusion profitable because $3,900 > 3,680$. (We subtract the fine from the profit in collusion. This is what the firms have to suffer when colluding.)
- 2. Levi's and Diesel are producing jeans. Each of them produce slightly different types of jeans - Levi's produces skinny jeans whereas Diesel focuses more on straight jeans. The market demand for Levi's jeans is $y_1 = 80 - p_1 + \frac{1}{2}p_2$, while the market demand for Diesel's jeans is $y_2 = 160 - p_2 + \frac{1}{2}p_1$. Levi's cost function is $TC_1(y_1) = 80y_1$ while Diesel's cost function is $TC_2(y_2) = 160y_2$.
 - Calculate the Bertrand equilibrium in this market. Indicate each firm's price, output level, and profits.
A: $p_1 = 128$, $p_2 = 192$, $y_1 = 48$, $y_2 = 32$, $\Pi_1 = 2304$, $\Pi_2 = 1024$
 - Find prices and output levels that would maximize joint profits, and calculate the maximum joint profits.
A: $p_1 = 146,7$, $p_2 = 213,3$, $y_1 = 40$, $y_2 = 20$, $\Pi_1 = 2666,7$, $\Pi_2 = 1066,7$ (all approximate)